**Question 1 (25 marks, CLO1, PLO1, C3)**

(a) Find a power series representation for $f(x)=\dfrac{12x^2}{1+x^3}$ and determine its interval of convergence. [5 marks]

(b) Determine whether the series $\sum_{n=1}^{\infty}\dfrac{2+\sqrt{3n}}{\sqrt{n^3+n+1}}$ is convergent or divergent. [5 marks]

(c) Consider the series $\sum_{n=1}^{\infty}\dfrac{1}{2^{n+\sin n}}$. Determine whether the series convergent or divergent.

[5 marks]

(d) Solve $\lim_{x\to 0}\dfrac{1-\cos x}{x\sin x}$. [5 marks]

(e) By writing the recurring decimal $0.\overline{12}=0.121212...$ as $\dfrac{12}{10^2}+\dfrac{12}{10^4}+...$, express $0.\overline{12}$ as a rational number in its simplest form. [5 marks]

**Question 2 (25 marks, CLO2, PLO1, C3)**

(a) Let $\vec{r}(t)=\left\langle\dfrac{\sin t}{t},t^2-3t+3,\cos t\right\rangle$. Find $\lim_{t\to 0}\vec{r}(t)$. [5 marks]

(b) Let $\vec{r}(t)=\langle t,t^2,t^3\rangle$ on $[-1.5,1.5]$. Find the vector equation of the line tangent to the graph of $\vec{r}$ at $t=-1$. [5 marks]

(c) Find the function $\vec{r}(t)$ with the properties:

$\vec{r}\,''(t)=\langle 2,\cos t,12t\rangle$,
$\vec{r}(0)=\langle -7,-1,2\rangle$, and

$\vec{r}\,'(0)=\langle 5,3,0\rangle$ [10 marks]

(d) Let $\vec{r}(t)=\left\langle\dfrac{\sin t}{t},t^2-3t+3,\cos t\right\rangle$. Determine whether $\vec{r}$ is continuous at $t=0$ and $t=1$.

[5 marks]

**Question 3 (25 marks, CLO2, PLO1, C3)**

(a) Find the nature of the critical points for the function $f(x,y) = x^3 + xy + y^2 - x$.

[20 marks]

(b) Let $f(x,y) = y\tan x + x\sec y$. Show that $f_{xy} = f_{yx}$. [5 marks]

**Question 4 (25 marks, CLO3, PLO2, C4)**

(a) Evaluate $\iiint (x+y+z)\,dxdydz$ over the tetrahedron which is bounded by the planes: $x=0,\ y=0,\ z=0$ and $x+y+z=1$. [10 marks]

(b) Find the volume of the solid bounded by the cylinder $x^2 + y^2 = 4$ and the planes $y+z=4,\ z=0$. [7 marks]

(c) Use the double integration to find the area lying inside the circle $r = a\sin\theta$ and outside the cardioid $r = a(1-\cos\theta)$. [8 marks]

**- END OF PAPER -**